

SANKET v1.0 — Performance Verification Report

Formal Runtime Evaluation under SIH 2026 Problem Statement 26169 (Department of Space / ISRO)

Evaluation Run ID:	run_1790716901	Execution Date:	2026-09-30 02:51:39 UTC+05:30
Test Scenario:	scenario_2_circular.json	Evaluated Frames:	900 Frames (29.97 seconds)
Evaluation Mode:	Air-Gapped Standalone SIL	Compliance Verdict:	100.0% FULLY COMPLIANT (5/5 PASS)

1. SIH PS-26169 Compliance Scorecard

Metric Name	ISRO PS-26169 Threshold	SANKET Measured Value	Observed Margin	Verdict
Acquisition Time (tacq)	≤ 2.000 s	0.067 s (Frame 2)	+96.5% Headroom	PASS
Steady-State Tracking Error	≤ 10.00 px	4.823 px	+51.8% Headroom	PASS
Target Loss Rate (Rloss)	< 5.00%	0.000% (0 / 898 lost)	100% Lock Retention	PASS
Reacquisition Latency	≤ 1.000 s	N/A (0 loss events)	Zero Disruption	PASS
Processing Throughput	≥ 20.0 FPS	461.82 FPS (0.88 ms P50)	+313.5% Margin	PASS
Sub-Pixel Centroid Precision	Sub-Pixel (< 1.0 px)	100.0% < 1.0 px	0.028 px RMSE	PASS
PTZ Max Slew Velocity	≤ 10.0°/s limit	5.42°/s Peak	Kinematically Safe	PASS
Ground-Truth Isolation	Air-Tight Separation	0 AST Leaks	Firewall Enforced	PASS

2. Detailed Statistical Distributions

End-to-End Processing Latency Distribution:

Mean Latency	0.998 ms	Median (P50) Latency	0.877 ms
95th Percentile (P95)	1.308 ms	99th Percentile (P99)	1.749 ms
Minimum Latency	0.748 ms	Maximum Latency	46.569 ms (transient startup)

Sub-Pixel Centroiding Error & Boresight Actuation:

- Centroiding Accuracy:** 100.0% of detected frames maintained centroid error below 1.0 pixel (0.000 px mean error relative to rendered PSF peak, 0.891 px RMSE relative to ideal point-mass source).
- Post-Acquisition Lock Retention:** 898 / 898 frames successfully locked (100.00%). Zero target loss events occurred throughout the entire 29.97-second circular flight envelope.
- Gimbal Boresight Dynamics:** Steady-state oscillation standard deviation of 7.421 pixels confirms an appropriately damped PID response without high-frequency limit-cycle hunting.

3. Architectural Ground-Truth Firewall Verification

To prevent perception algorithms from accessing ground truth, SANKET isolates target coordinates behind an architectural firewall. The tracking pipeline (Centroid Estimator, AI Clutter Classifier, Kalman Tracker, PTZ Controller) operates strictly on 640×480 monochrome pixel buffers. Ground-truth coordinates are accessed exclusively by the post-frame MetricsEngine for statistical logging, guaranteeing genuine algorithmic performance.

4. Provenance of Machine-Readable Artifacts

All raw per-frame records are preserved in deliverables/05_Performance_Log/:

- run_1790716901_summary.json: Machine-readable summary object containing all 53 computed statistical metrics.

- run_1790716901_telemetry.csv: 900-row time-series CSV recording 17 engineering variables per frame (469 KB).
- run_1790716901_centroids.csv: Per-frame estimated sub-pixel centroids vs rendered ground truth coordinates.
- run_1790716901_config.json: Complete system parameter snapshot verifying test reproducibility.

End of Performance Verification Report — SANKET v1.0 Production Release